Comfort Thermostat

How our AC is controlled, and what ten days of temperature data show about the flat

Jerusalem · written 21 October 2026 · data 11–21 October 2026

In short.

- Home Assistant now runs the ducted AC as a thermostat. It watches one room at a time and cools in short bursts when that room gets more than half a degree above its comfort temperature.
- Since today it is set to **21.5 °C in the bedroom at night and 23 °C in the living room by day**, one degree warmer than before.
- It uses gentler, shorter bursts than at the start of the week, and it never uses the AC's fan-only mode.
- The flat itself is very stable. It sits around **20.5–21.5 °C** whatever the weather does outside. The nursery is consistently the coolest room, and the bedroom heats up in the afternoon sun.
- At this time of year the house is often already below our targets without any help, so the AC will mostly rest.

1. How the AC is controlled

The AC is the central Electra ducted unit. It has no network connection, so Home Assistant talks to it the same way the handheld remote does: an infrared blaster (Broadlink RM4C) sends it remote-control signals. We decoded the remote's signal format in July and generated a code for every combination of mode, temperature and fan speed.

Infrared only goes one way, and that has two consequences:

- **Home Assistant can't see what the AC is actually doing.** It keeps an *assumed* state: on or off, mode, setpoint, fan. If someone uses the handheld remote, the two can drift apart until Home Assistant next sends a full command.
- **Power is a toggle.** There is no separate "on" and "off" signal, so the thermostat never switches the unit on or off. It only changes mode, temperature and fan while the AC is (assumed to be) on.

The AC only ever runs in **cool** (or heat, manually). **Fan-only and dry modes are never used.** The control doesn't offer them and there are no codes for them. If the unit is ever found in fan mode, it was set accidentally from the handheld remote. The next command from Home Assistant puts it back to cool.

2. How the thermostat decides

The thermostat follows **one room at a time**: the bedroom or the living room. The office has no temperature sensor yet. With **auto room** switched on, it follows the bedroom from 22:00 and the living room from 09:00. Each room has its own comfort temperature, and in practice these two are our night and day settings.

Every five minutes it compares that room with its target and puts the AC in one of two states:

Cooling	Starts when the room is **0.5 °C or more above** the target. The AC is set **1 °C below** the target with the fan on low. It goes to medium only when the room is a full degree over, and high at two degrees over. When the room is 1.5 °C or more over, the setpoint goes 2 °C below the target.
Resting	The AC is set to 28 °C, which keeps the compressor off in cool mode, with the fan on low.

It stops cooling **before** the room reaches the target, because the ducts keep blowing cold air for several minutes after the compressor stops. Each check estimates where the room will be in ten minutes (from how fast it is cooling). If that estimate is within 0.2 °C of the target, it switches to resting.

Settings right now

Comfort temperature	°C
Bedroom (night, from 22:00)	21.5
Living room (day, from 09:00)	23.0
Office (no sensor yet)	23.0

Tuning setting	Value
Start cooling at	+0.5 °C
Stop cooling at	0 °C
Stop early if forecast within	+0.2 °C
Forecast looks ahead	10 min
Setpoint while cooling	target – 1
Fan medium / high at	+1 / +2 °C

You can change the comfort temperatures on the Living Room and HVAC dashboards. The tuning settings are meant to be changed by the self-tuning routine (section 5), not by hand.

3. How it got here

When	What changed and why
Jul 2026	Decoded the remote's signal format. The AC became controllable from Home Assistant with on/off, mode, temperature and fan.
22 Jul	First "Climate Manager": fixed night and day temperature limits.
19 Oct, morning	Replaced by the Comfort Thermostat (v1). It nudged the AC's setpoint down 1 °C every ten minutes. **Problem:** the flat cools about 1 °C every ten minutes once the AC runs, so it wound the AC down to 17 °C before the room had responded, then held it there. The living room was freezing.
19 Oct, 14:15	v2: switched to the cooling / resting cycle with the ten-minute forecast. No more wind-down to 17 °C.
21 Oct	v2.1: **gentler.** v2's bursts (AC at 20 °C, fan medium) still took the living room about 0.3 °C below target each time, which felt like a cold blast. The setpoint is now only 1 °C below target, the fan stays on low longer, and it stops earlier.
21 Oct	**+1 °C on all comfort temperatures** (bedroom 20.5 → 21.5, living room 22 → 23). Removed an old, broken AC control that still offered fan-only mode.
21 Oct	The tuning settings became adjustable values, and a self-tuning routine was added (section 5).

4. What the data shows about the flat

The figures below cover **11–21 October**, the full history Home Assistant keeps (about ten days). They come from the temperature and humidity sensors in the bedroom, living room and nursery, plus the official outdoor reading for central Jerusalem.

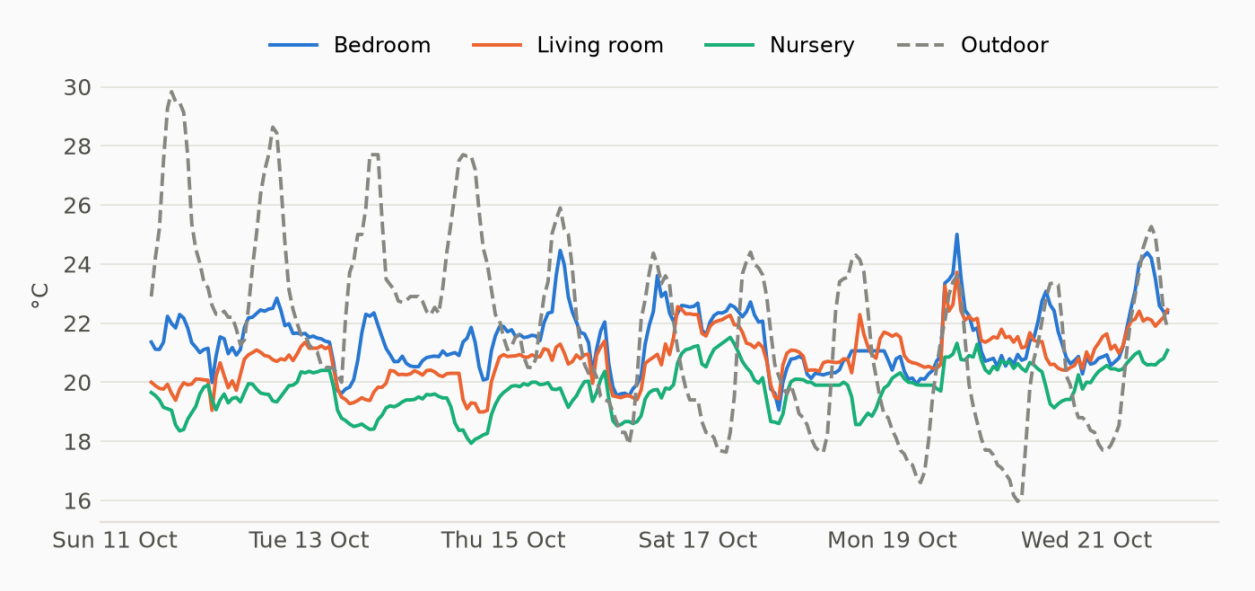


Figure 1: Hourly average temperature, 11–21 October. Outdoor is dashed.

The flat is very stable

Outside, the temperature swung from **15.9 °C to 29.9 °C**, often 8–10 °C within a single day. Inside, the living room averaged **20.8 °C** and spent most of the period between 20 and 22 °C. Overnight the bedroom and living room are almost identical (within 0.1 °C on average). The flat holds heat and coolness well. A stone building with ducted air does that, and it's the main reason the thermostat has little to do at this time of year.

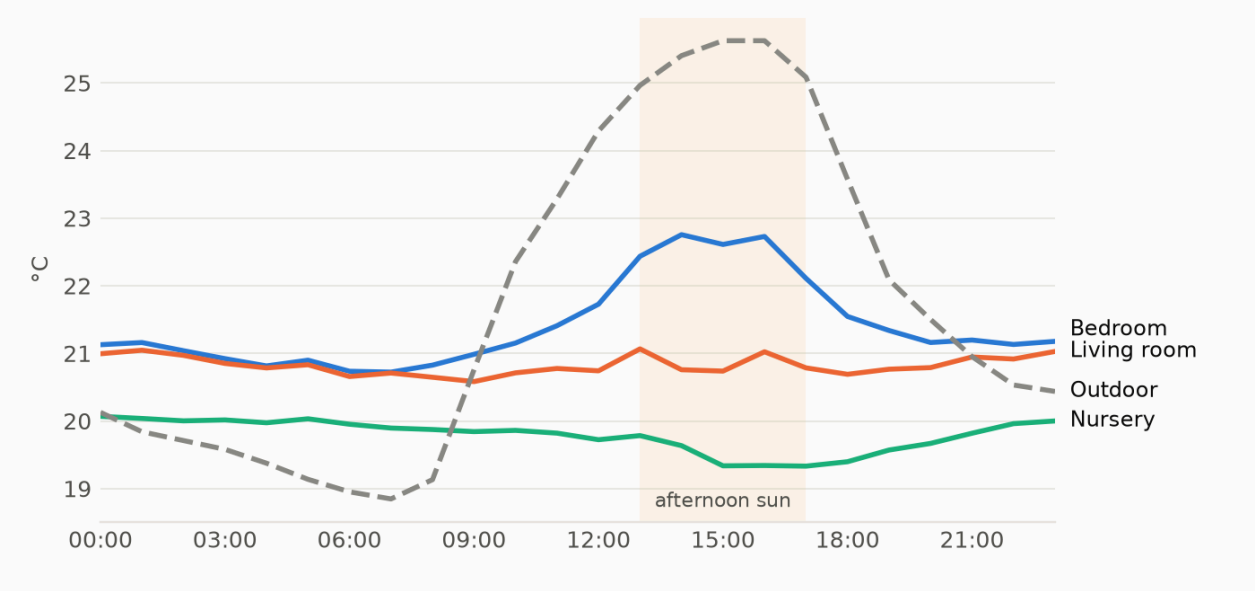


Figure 2: Average temperature by time of day across the ten days.

Room by room

Room	Average	Range	Humidity	Pattern
Bedroom	21.4 °C	18.9–25.9	65 %	Warms every afternoon: **+1.7 °C above the living room** between 13:00 and 17:00, peaking at about 22.7 °C on average and 24–26 °C on the warmer days (15, 16, 19 and 21 Oct). This is sun on that side of the flat.
Living room	20.8 °C	18.6–25.0	71 %	Flat all day, with only about 0.5 °C between night and afternoon.

Room	Average	Range	Humidity	Pattern
Nursery	19.8 °C	17.9–21.6	71 %	**The coolest room**, about 1 °C below the living room all day and 1.4 °C below in the afternoon. It gets slightly cooler in the afternoon, the opposite of the bedroom.
Outdoor	21.8 °C	15.9–29.9	—	Getting cooler through the period: daily highs fell from about 29 °C to about 24 °C, and nights from 22 °C to 16 °C.

Comfort

In July we set our comfort band at **19–21.5 °C**, adjusted for humidity: humid air feels warmer, so each 10 % of humidity above 50 % counts as +0.3 °C. Indoor humidity has been high, **65–71 %** on average, which pushes the "feels like" temperature up by about half a degree. On that measure:

Room	In band	Warmer	Cooler
Bedroom	39 %	61 %	0 %
Living room	52 %	48 %	0 %
Nursery	83 %	10 %	7 %

That doesn't match how the flat has **felt** this week. The living room at 22 °C felt too cold, which is why the targets went up today. Possible reasons:

- the July band was chosen in midsummer;
- the AC's air stream feels colder than the room temperature;
- the humidity correction overstates how warm it feels.

We should revisit the band rather than trust it.

What this means for the thermostat

- **Most of the cold in the living room was not the thermostat.** Since it started on 19 Oct, the living room was more than half a degree below its 22 °C target for about 39 % of daytime, and nearly all of that time the AC was *resting*. On 20 Oct the room sat at 20.3–21.5 °C all day without one cooling burst. The flat simply runs cooler than 22 °C in this weather.
- **Two things are still worth checking.** The "resting" state still runs the fan on low, and air moving through the ducts can feel cool. The AC may also not always be in the state Home Assistant assumes.
- **With the new targets (21.5 °C at night, 23 °C by day), the AC should rarely need to cool** until the weather warms again, except perhaps for the bedroom on sunny afternoons. The bedroom isn't the followed room by day, though.
- **Heating season is coming.** The thermostat only cools. If the flat drifts below about 20 °C, it will do nothing about it.

Caveats: Home Assistant's AC on/off state is assumed, not measured. The data before 13 Oct has no record of the AC's state at all, and the AC was probably running on its old schedule. Outdoor readings come from a weather station a few kilometres away, updated about hourly.

5. Self-tuning

The thermostat now has a self-tuning routine, a Claude skill kept in a plugin for our Home Assistant. When asked to "tune the thermostat", it:

1. reads the record of past changes and the data since the last one;
2. measures every cooling burst: how far the room dropped below target, how long it ran, and how often it repeated. It does this per room, by day and by night;
3. if there's enough evidence (at least five bursts in a room), moves **one** setting by **one small step** within safe limits, and records why.

It never changes the comfort temperatures (those are ours to choose), the AC's mode, or the code itself. It also reports problems that no setting can fix, such as the flat being colder than the target by itself. Tested on this week's data, it reached the same conclusion we reached by hand: stop the bursts earlier. It needs two to four days under today's settings before it can judge them.

6. Open questions

- **Is “resting” really resting?** If the fan on low feels draughty, the thermostat could switch the AC off when the room is well below target, instead of leaving it resting.
- **Heating.** Add a heating side before winter, or keep heating manual.
- **Office sensor.** The office can’t be followed until it has a temperature sensor.
- **Bedroom afternoon sun.** Blinds closed in the early afternoon would likely do more than the AC, which can’t cool the bedroom without over-cooling the rest of the flat.
- **Better AC feedback.** The planned replacement blaster (Seeed XIAO IR Mate) can also **hear** the handheld remote, so Home Assistant would know when someone changes the AC by hand.